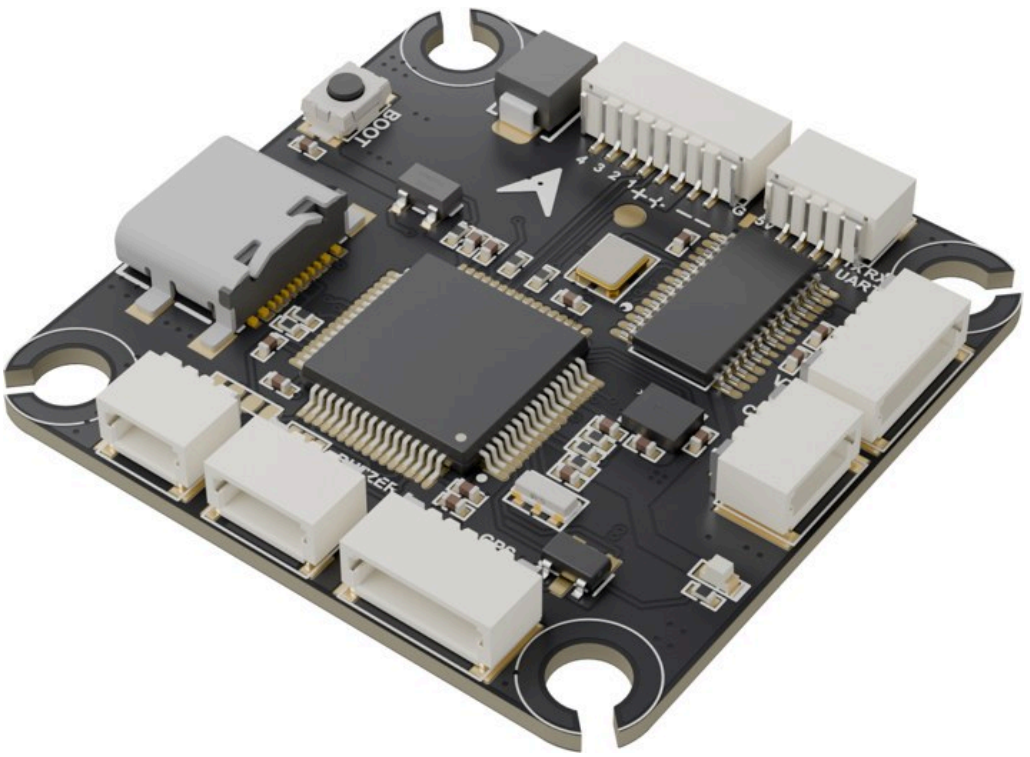
AF-F4 nano

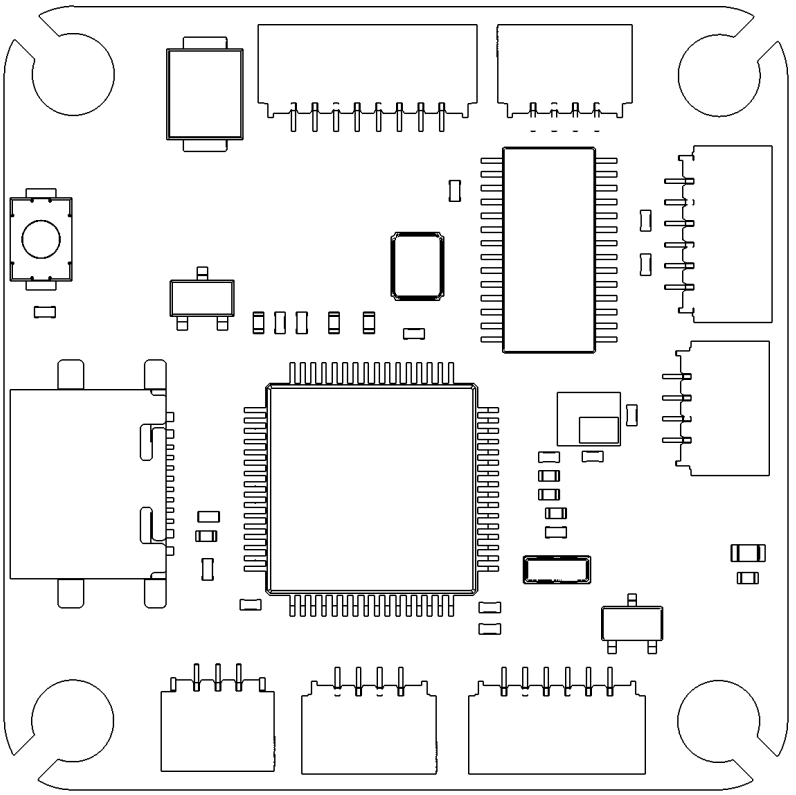
User Guide

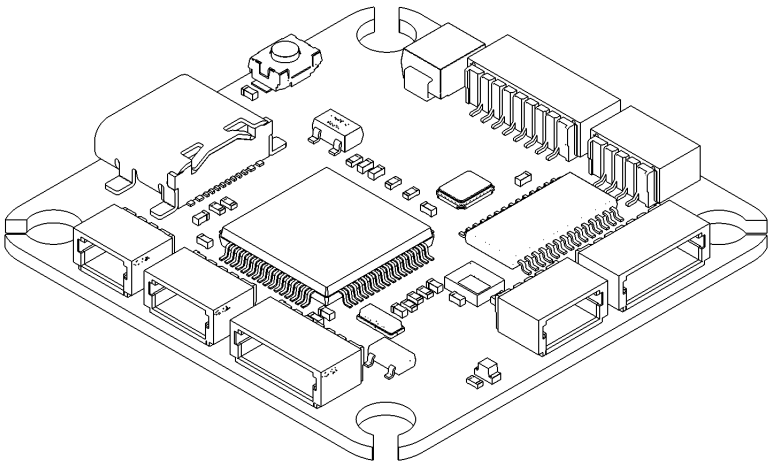


novaX

1. Parts List

· Main Board





AF-F4 nano * 1

1. Parts List

· Connectors & Pads

Port	Type
ESC	8-Pin · silk 4 3 2 1 + + − −
ELRS	4-Pin · UART4 · silk G 5V TX RX
VTX	6-Pin · UART3 TX · maker's order
CAM	3-Pin · video in · maker's order
RC	3-Pin · PPM · maker's order
BUZZER	4-Pin · UART3 RX · maker's order
GPS	6-Pin · UART1 + I2C1 · maker's order
SBUS pads	UART2 RX · RC input
RGB pads	UART3 / UART5 · LED or serial
I2C / UART pads	I2C1 / UART1
VTX pads	UART3 TX · VTX control
MOTOR / POWER pads	Motor signals S1–S8 + battery

2. Specifications

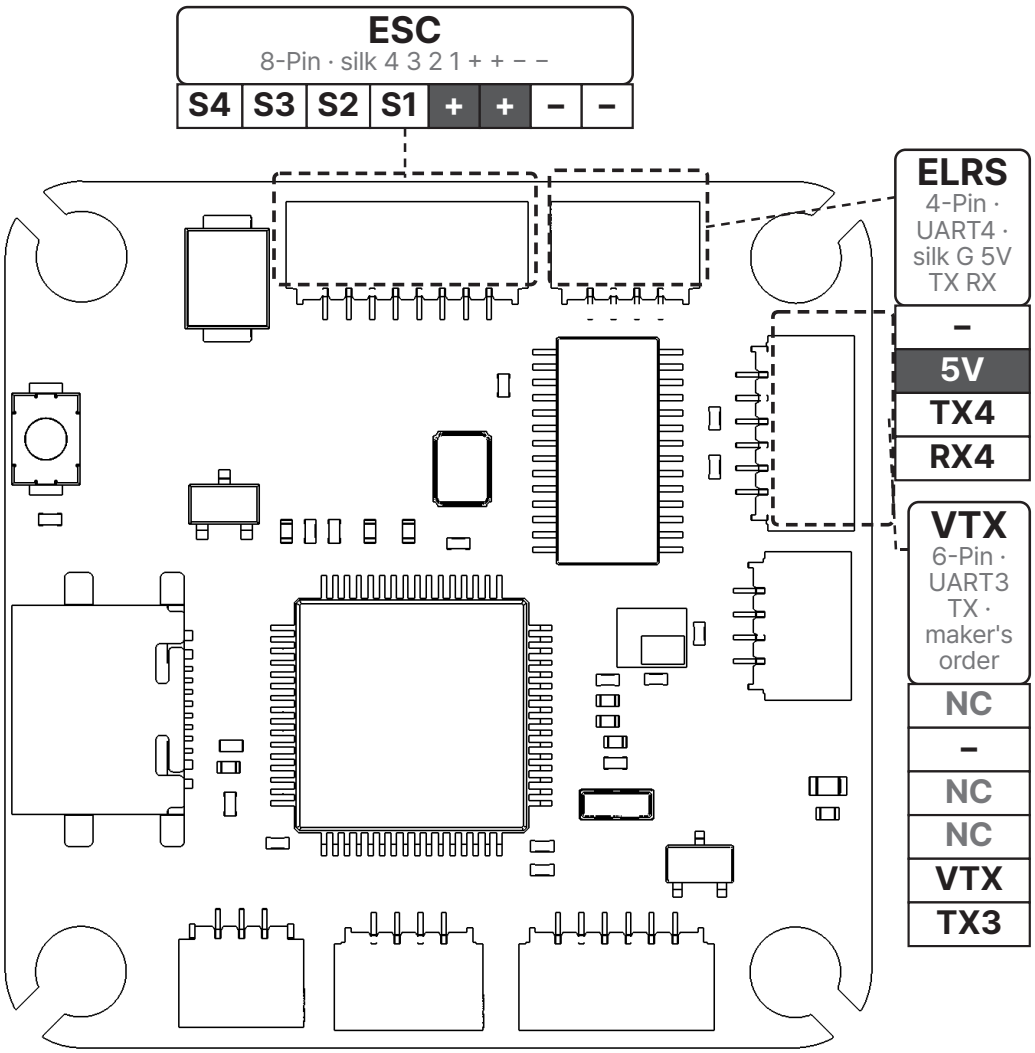
· Hardware Specifications

MCU	STM32F405
IMU	ICM-42688-P
Barometer	SPL06
Operating Voltage	9 – 25 V DC
Output Voltage	3.3V/1A · 5V/3A · 10V/3A
USB Input	USB Type-C
PWM Output	8 motor/servo channels + 1 LED channel
Serial Ports	6 hardware UARTs + USB configured; verify connector pinout
Servo Output	2 Port
RC Input	PWM / PPM / S.Bus
Size	39.4 × 39.4 mm
Mount Hole	30.5 × 30.5 mm / M4
Weight	9.3 g
Operating Temp	-20 ~ +70 °C
Supported F/W	novaX ArduPilot; Betaflight board configuration available

3. Pinouts

· Top (1)

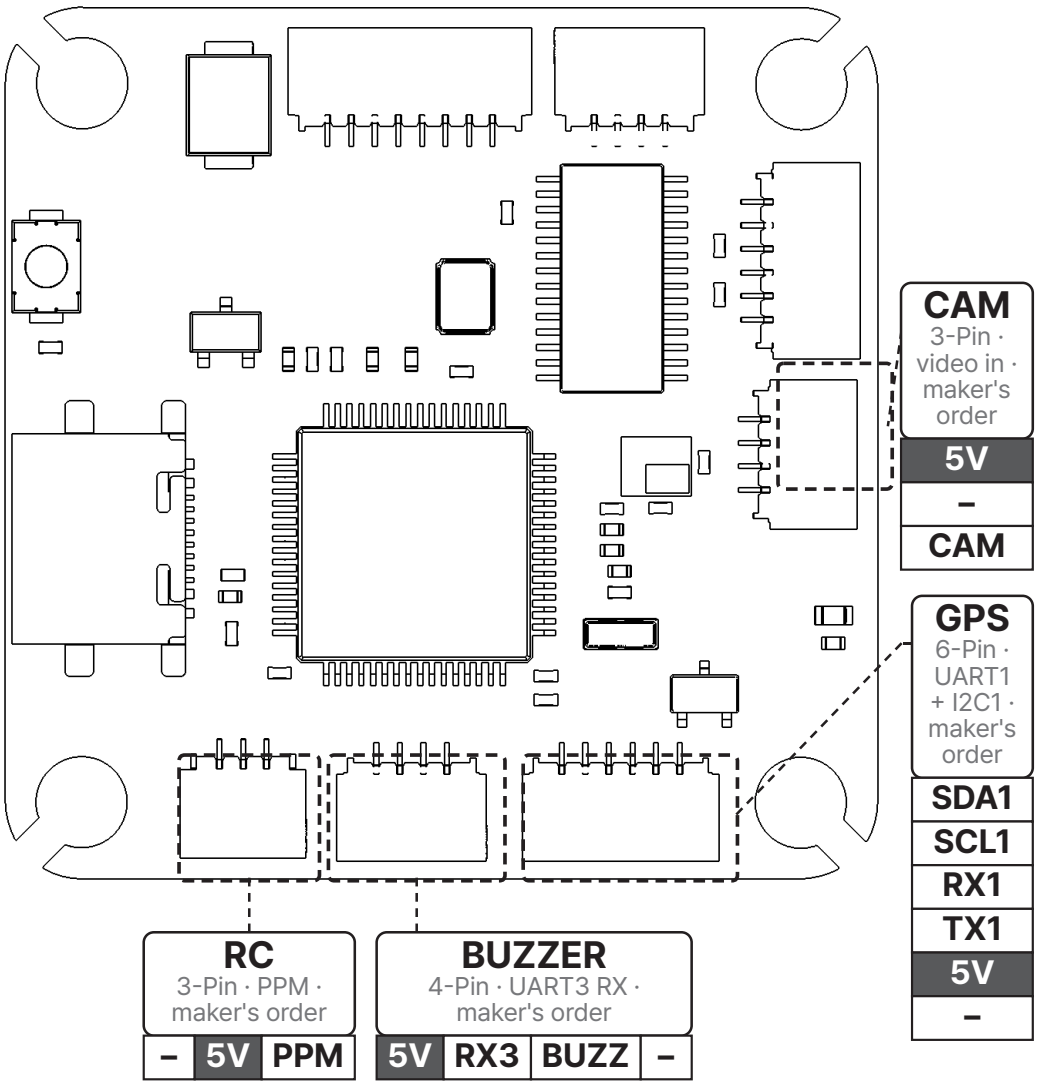
❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture



3. Pinouts

· Top (2)

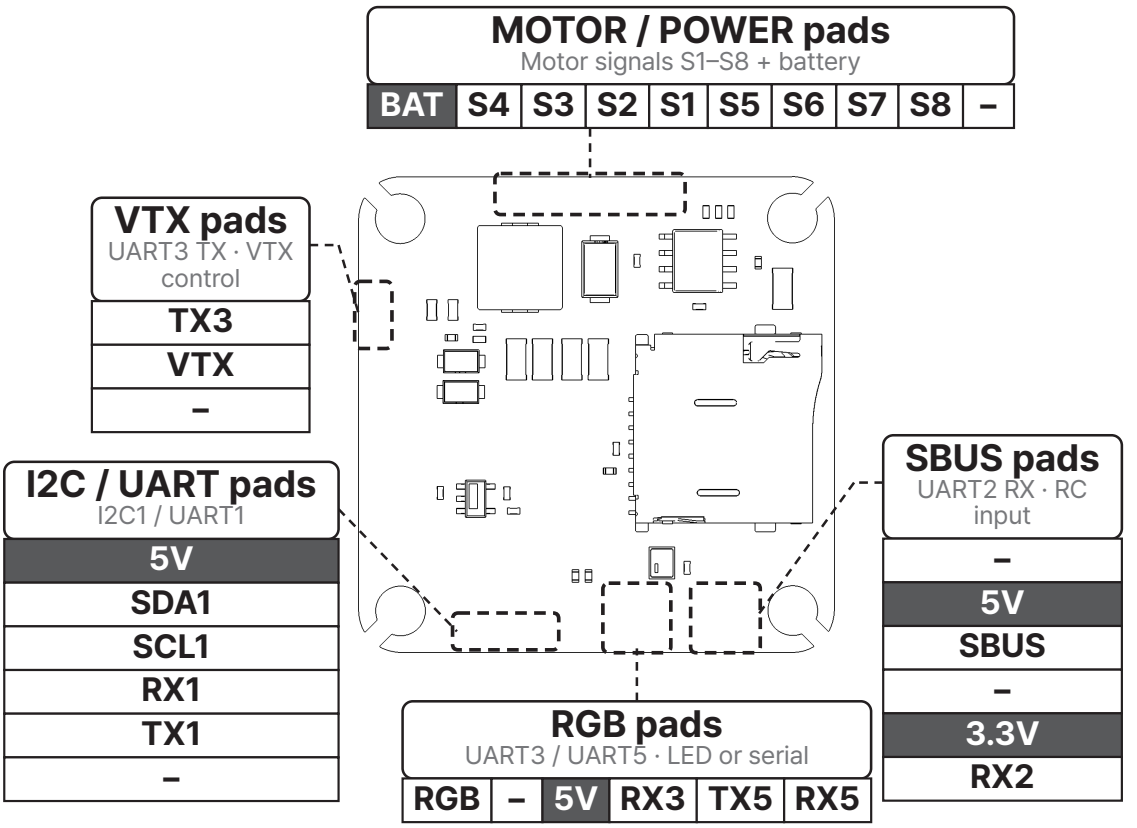
❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture



3. Pinouts

· Bottom

❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture



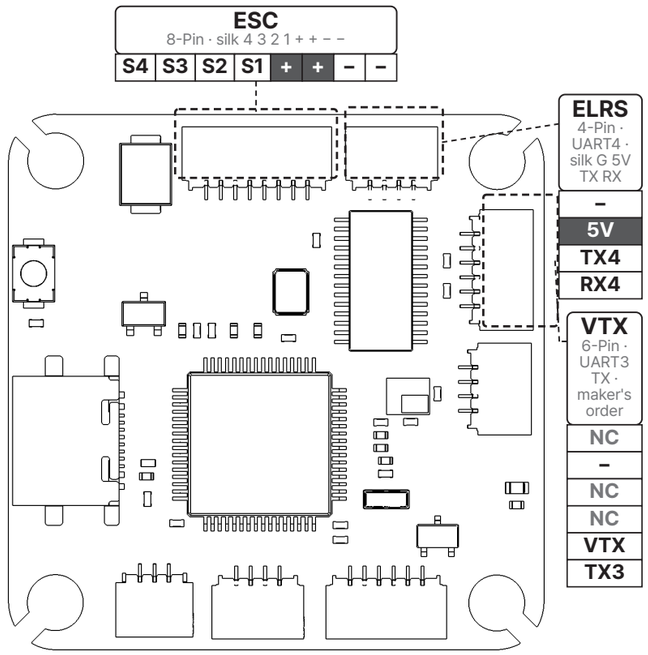
novaX

AF-F4 nano

3. Pinouts

· Top (1)

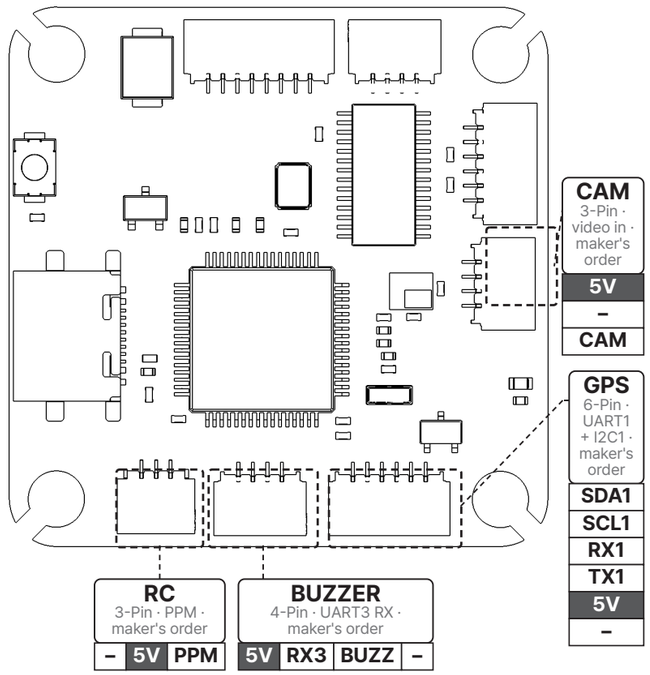
❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture



3. Pinouts

· Top (2)

❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture



3. Pinouts

· Bottom

❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture

